

HDOC DAY-30

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Batch-15

Question:

Write a menu-driven C program to perform the following operations on a given array:

1. Sort the array using Bubble Sort and perform Binary Search.
2. Insert and delete an element at a valid position.
3. Rotate the array elements to the right by k positions.
4. Exit the program.

Aim

To write a **menu-driven C program** to perform Bubble Sort with Binary Search, insertion, deletion, and right rotation on a given array.

Algorithm

1. Start.
2. Read the size of the array and its elements.
3. Display the menu:
 - Bubble Sort and Binary Search
 - Insertion and Deletion
 - Right Rotation by k positions
 - Exit
4. Read the user's choice.
5. Perform the selected array operation.
6. Display the resulting array or search result.
7. Repeat the menu until **Exit** is selected.
8. Stop.

>C PROGRAM:

```
1 #include <stdio.h>
2 int main()
3 {
4     int a[100], m, ch, i, j, t, key, l, h, m, pos, s, k;
5     printf("Enter size: "); scanf("%d", &n);
6     printf("Enter elements: ");
7     for(i=0; i<n; i++) scanf("%d", &a[i]);
8
9     printf("\n 1.Sort 2.Search 3.Insert 4.Delete 5.Right Rotate 6.Exit\nChoice: ");
10    scanf("%d", &ch);
11
12    switch(ch)
13    {
14        case 1:
15            for(i=0; i<n-1; i++)
16            {
17                for(j=i+1; j<n; j++)
18                {
19                    if(a[j]<a[i])
20                    {
21                        t=a[i]; a[i]=a[j]; a[j]=t;
22                    }
23                }
24            }
25            printf("Sorted "); for(i=0; i<n; i++) printf("%d ", a[i]);
26            printf("\nSearch: "); scanf("%d", &key);
27            l=0; h=n-1;
28            while(l<=h)
29            {
30                m=(l+h)/2;
31                if(a[m]==key) break;
32                if(a[m]<key) l=m+1; else h=m-1;
33            }
34            if(l>h) printf("Found at %d", m); else printf("Not found");
35            break;
36
37        case 2:
38            printf("Position & value: "); scanf("%d%d", &pos, &val);
39            for(i=pos; i<n; i++) a[i]=a[i+1];
40            a[pos]=val;
41            printf("Delete position: "); scanf("%d", &pos);
42            for(i=pos-1; i<n-1; i++) a[i]=a[i+1];
43            a[pos]=0;
44            printf("Array: "); for(i=0; i<n; i++) printf("%d ", a[i]);
45            break;
46
47        case 3:
48            printf("Enter k: "); scanf("%d", &k);
49            while(k>0)
50            {
51                t=a[n-1];
52                for(i=n-1; i>0; i--) a[i]=a[i-1];
53                a[0]=t;
54            }
55            printf("Array: "); for(i=0; i<n; i++) printf("%d ", a[i]);
56            break;
57
58        case 4:
59            printf("Invalid choice");
60    }
61    return 0;
62 }
```

>TEST CASE TABLE:

Case	Input	Expected Output
1. Sort & Search	40 10 30 20 50, key=30	10 20 30 40 50, Found
2. Insert & Delete	Insert 25 at 3, Delete 2	Updated array
3. Right Rotate	10 20 30 40 50, k=2	40 50 10 20 30
4. Exit	Choice = 4	Program ends

>OUTPUT:

```
Session / Terminal Emulator Use the command line
zsh: CORTW...
priyadarshna@kali:~$ gcc day30.c -o day30
priyadarshna@kali:~$ ./day30
Enter size: 5
Enter elements: 40 10 30 20 50
1.Sort & Search
2.Insert & Delete
3.Right Rotate
4.Exit
Choice: 1
Sorted: 10 20 30 40 50
Search: 30
Found at 3
1.Sort & Search
2.Insert & Delete
3.Right Rotate
4.Exit
Choice: 2
Position & value: 3 25
Delete position: 2
Array: 10 25 30 40 50
1.Sort & Search
2.Insert & Delete
3.Right Rotate
4.Exit
Choice: 3
Enter k: 2
Array: 40 50 10 25 30
1.Sort & Search
2.Insert & Delete
3.Right Rotate
4.Exit
Choice: 4
priyadarshna@kali:~$
```